DVST

DVST (Direct View Storage Tube) is a type of CRT display that **stores the image on the screen**, so it **does not require continuous refreshing**. It uses **two electron guns**: a primary gun to store the image and a flood gun to maintain it.

Refresh CRT

Refresh CRT is a type of CRT display that **continuously redraws (refreshes) the image** on the screen, usually **60–80 times per second**, to keep the picture visible and prevent it from disappearing.

Feature	DVST (Direct View Storage Tube)	Refresh CRT
Definition	Stores the image on the screen, so refreshing is not required.	Continuously redraws the image to keep it visible.
Refreshing	No continuous refreshing is needed.	Requires continuous refreshing (about 60–80 frames/sec).
Electron Guns	Uses **two electron guns** (Primary gun and Flood gun).	Uses **one electron gun**.
Image Storage	Image is stored as a **charge pattern** behind the phosphor screen.	Image is stored in the **frame buffer** and refreshed repeatedly.
Flickering	No flicker because the image remains on the screen.	Flicker may occur if the refresh rate is low.
Image Erasing	Individual parts of the image **cannot be erased**.	Images can be **changed or erased easily**.
Animation	Not suitable for animation.	Suitable for animation and moving graphics.
Resolution	Can display **high-resolution complex images**.	Resolution depends on the refresh system and display hardware.
Memory Requirement	Does not require a large refresh memory.	Requires a **frame buffer (refresh memory)**.
Applications	Used for **static engineering drawings and CAD**.	Used in **computer monitors and television displays**.

DVST	Refresh CRT
D = Directly stores image	**R = Refreshes image continuously**
No refresh needed	Refresh needed
Two electron guns	One electron gun
No flicker	Flicker possible
No animation	Good for animation